SpeechTech Week 7 AQ PA GA

1) Sate True or False: The observation probability for the weather prediction HMM model is an example of continuous HMM model. *1 point*

→ discrete

- ○ True
- ● False

lexicons

2) Once the best sequence of corresponding phonemes from the observed MFCC features is found using viterbi, ______ is used for finding the corresponding word.

lexicon

1 point

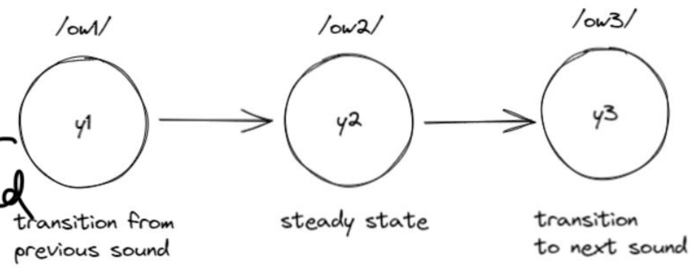
1) The tri-phone model uses three separate models based on ____ and ____ context.

left, right

L R

1 point

2) For the three state model as shown in figure, each state is modelled by _____ *1 point*



modelled using GMM

- ● GMM
- ○ HMM

write speak

1) In most Indian languages we _____ what we _____ ,making it easier to train language models.

write, speak

1) Language models help a speech recogniser figure out how likely a word sequence is, independent of ____ *1 point*

acoustic

- ● acoustics
- ○ words
- ○ grammar
- ○ None of these

2) For a sentence having three words $(w1, w2, w3)$, the probability of the word sequence $P(w1, w2, w3)$ can be expanded as follows (hint: use Bayes' rule) *1 point*

- ○ $P(w1)P(w2)P(w3)$
- ○ $P(w1) + P(w2) + P(w3)$
- ● $P(w1)P(w2|w1)P(w3|w2, w1)$
- ○ None of these

$P(w_1) \times P(w_2|w_1) \times P(w_3|w_1, w_2)$

3) Where are Language Models used? *1 point*

- ○ Text Generation
- ○ Next Word Prediction
- ○ Speech Recognition
- ● All of these

most probable word seq / speech

1) Which of the following can be the definition of ASR problem? **1 point**

- Finding most probable speech sequence given the word sequence
- Finding most probable word sequence given the speech sequence
- Finding most probable word sequence given the text sequence

2) State whether the following statement is true or false: N-gram is defined as a sequence of n tokens chosen randomly from a text. **1 point**

α

- True
- False

3) State whether the following statement is true or false: N-gram Language Model is able to capture long range dependencies and the context very well without any difficulty. **1 point**

α

- True
- False

1) ______ act as inputs to the biological neurons **1 point**

dendrites

- Nucleus
- Axon
- Dendrite
- Cell body

→ axon

2) State whether the following statement is true or false: Synapses in biological neurons are always excitatory in nature. **1 point**

- True
- False

excitatory or inhibitory

1) State whether the following statement is true or false: In artificial neural networks, all inputs to a neuron are associated with the same weight. **1 point**

α

- True
- False

2) Which Boolean function is implemented by the following perceptron function? **1 point**

$\Sigma_{i=1}^{2} x_i \geq 1$

- AND
- OR
- XOR
- NOR

≥ 1

$$\left.\begin{array}{lll} 0 & 0 & \Rightarrow 0 \\ 0 & 1 & \Rightarrow 1 \\ 1 & 0 & \Rightarrow 1 \\ 1 & 1 & \Rightarrow 2 \end{array}\right\} 1$$

3) Which Boolean function is implemented by the following perceptron function? **1 point**

$\Sigma_{i=1}^{2} x_i \geq 2$ =

- AND
- OR

4) Which of the following Boolean functions is a non linearly separable function? **1 point**

XOR

- AND
- OR
- XOR
- NOR

1) State whether the following statement is true or false: The perceptron function behaves like a step function. **1 point**

- True
- False

2) If $f(z) = RELU(z)$, then the derivative of $f(z)$, when $z \geq 0$, is? **1 point**

$$f(z)' = \begin{cases} 0 & z < 0 \\ 1 & z > 0 \end{cases}$$

- 0
- 1
- z
- None of the above

3) Which of the following is a sigmoid function? **1 point**

$$\frac{1}{1+e^{-z}}$$

- e^{-x}
- $\frac{1}{1+e^{-x}}$
- e^{x}

1) Sate True or False: The observation probability for the speech recognition HMM model is an example of continuous HMM model. **1 point**

- True
- False

2) For better speech recognition and to handle the transition between phonemes, each phoneme is modelled by how many states? **1 point**

3

- 1
- 4
- 16
- 3

3) In an N-gram language model, the probability of occurrence of a word, ________ **1 point**

- depends on the previous word only
- depends on the previous N words
- depends on the previous N-1 words
- does not depend on the previous words

4) The ASR problem can be defined as `Finding the most probable word sequence (W) given as a speech sequence(S)', we encounter $P(S|W)$ while solving this problem. What does it represent? **1 point**

- Language Model
- Hidden Model
- Acoustic Model
- None of these

$P(s|W)\,P(W)$

acoustic model

$P(s)$

lang. model

5) The biological neurons send output through ______ **1 point**

- Nucleus
- Axon
- Dendrite
- Cell body

→ axon

6) Which of the following Boolean functions cannot be implemented by a single perceptron? **1 point**

- AND
- OR
- XOR
- NOR

XOR

7) If $f(z) = tanh(z)$, then the derivative of $f(z)$ will be? **1 point**

- $f(z)$
- $1 - f(z)$
- $1 - f^2(z)$
- $f^2(z)$

$f(z) = \tanh(z)$

$f(z)' = 1 - \tanh^2(z)$

8) If $f(z) = RELU(z)$, then the derivative of $f(z)$, when $z < 0$, is? **1 point**

- 0

0

9) For the word rat, the phoneme expansion can be written as /r/, /a/, /t/. The correct left context short-hand for /a/ is written as: **1 point**

- r_+
- r_-
- t_+
- None of these

a

r_- t_+

1) Suppose that in the training corpus, the word sequences "you" occurred 500 times, "How are you" occurred 300 times and "How are" occurred 800 times, what is the probability that the word "you" follows the word sequence "How are"? ***1 point***

you – 500

how are you → 300

how are — 800

- 0.3
- 0.625
- 0.8
- 0.375

$P(\text{you} \mid \text{how are}) = \frac{P(\text{how are} \mid \text{you})\, P(\text{you})}{P(\text{how are})}$

$= \frac{P(\text{how are} \cap \text{you})}{P(\text{how are})} = \frac{300}{800} = 0.375$

No, the answer is incorrect.
Score: 0

Accepted Answers:

0.375

2) Suppose you are given two language models A and B. You compute the perplexity with respect to a test set for both models and you find out that $\text{perplexity}(A) < \text{perplexity}(B)$. Which model do you think is better? ***1 point***

- A
- B

A > B p↓

3) Which Boolean function is implemented by the following perceptron function?, ***1 point***

$\Sigma_{i=1}^{2}(-x_i) \geq 0$

- AND
- OR
- XOR
- NOR

Yes, the answer is correct.
Score: 1

Accepted Answers:

NOR

4) Which of the following Boolean functions cannot be implemented by a multilayer perceptron? ***1 point***

- AND
- OR
- XOR
- None of the above

5) If $f(z) = \frac{1}{1+e^{-z}}$, then the derivative of $f(z)$ will be? **1 point**

- $f(z)$
- $1 - f(z)$
- $f(z)f(1-z)$
- $f(z)(1-f(z))$

$f(z)' = (1 - f(z))\, f(z)$

Yes, the answer is correct.
Score: 1

Accepted Answers:

$f(z)(1-f(z))$

6) Select the correct relationship between $P(x^{(1)}, ..., x^{(T)})$ and $P(x^{(t)}|x^{(t-1)}, ..., x^{(1)})$ **1 point**

- $P(x^{(1)}, ..., x^{(T)}) = P(x^{(T)}|x^{(T-1)}, ..., x^{(1)})$
- $P(x^{(1)}, ..., x^{(T)}) = \prod_{t=1}^{T} P(x^{(t)}|x^{(t-1)}, ..., x^{(1)})$
- $P(x^{(1)}, ..., x^{(T)}) = \sum_{t=1}^{T} P(x^{(t)}|x^{(t-1)}, ..., x^{(1)})$
- None of these

7) Consider a language with a total of 50 phonemes, how many separate models for a single phoneme will need to be created considering a tri-phone model setup? **1 point**

- 2401
- 2500
- 3
- None of these

$2500 - (5\cancel{0}0 + 50)$

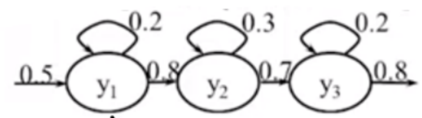
$= 2400$

No, the answer is incorrect.
Score: 0

Accepted Answers:

2401

8) The given image represents which of the following speech model. **1 point**



mono

- 3-state triphone
- 3-state monophone
- 1-state triphone
- 1-state monophone

tri